Entomological Law

A Field Map of Insects as Evidence, Threat, Property, Product, Patient, and Weapon

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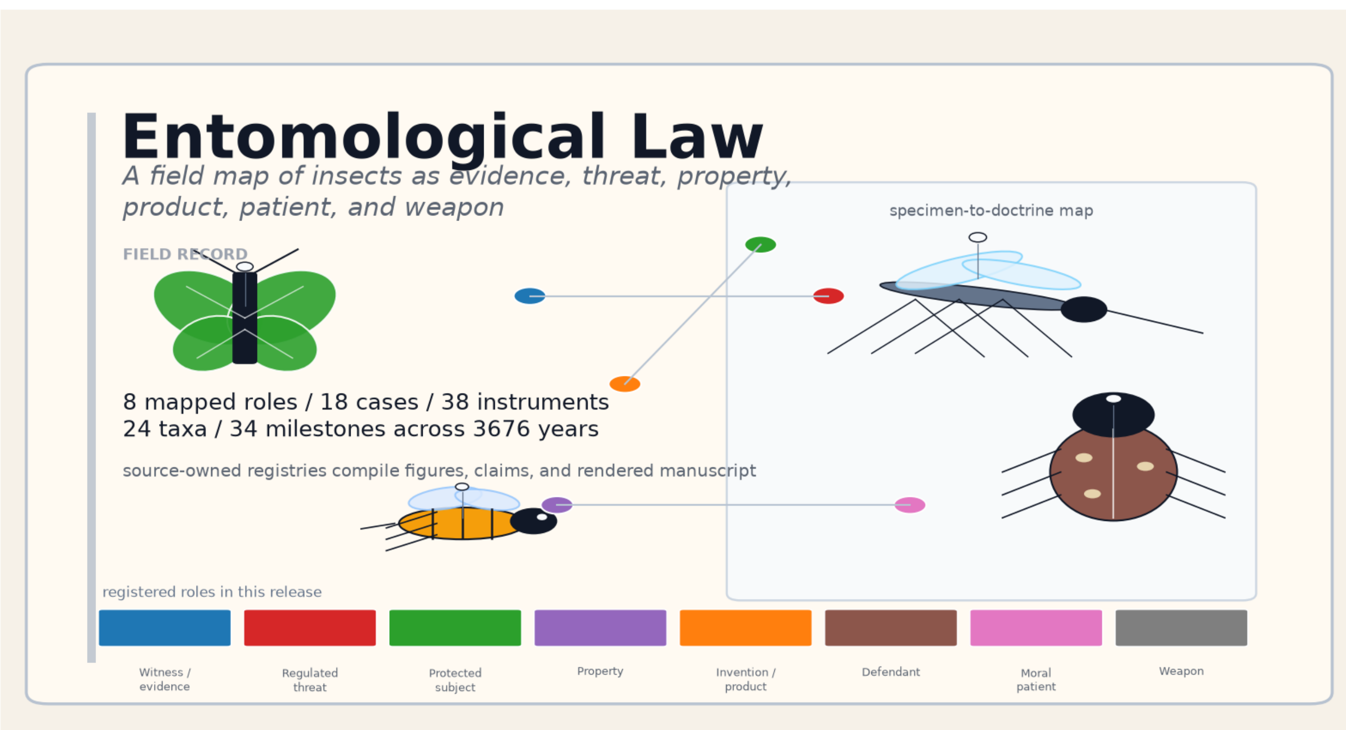

Contents

1 Abstract: A Source-Anchored Map of Entomological Law

There is no statute, treatise, or law-school casebook titled "Entomological Law." The phrase names a *synthetic field* — the convergence zone where the six-legged world repeatedly forces the legal system to answer questions it was not designed for: *Can a fly testify? Who owns a swarm? Is a bumblebee a fish? Can you patent a mosquito? May a court excommunicate a weevil? Does a cricket suffer? May insects be used in war?* This reference compiles that field as a machine-readable and reproducible artifact organized around the legal *role* an insect occupies in a given dispute. It encodes 8 legal roles — witness, regulated threat, protected subject, property, invention, defendant, moral patient, and weapon — and binds to them 18 landmark decisions, 38 statutes and treaties across 9 categories and 7 jurisdictions, 24 insect taxa, 12 certifying and regulatory institutions, 34 historical milestones spanning 3676 years, and 5 cross-domain themes that knit the roles together. Every count in this prose is generated from the source registries under `src/`, legal propositions are source-bound in the bibliography, every externally-sourced statistic written as a numeral is bound to a verification record in the claim ledger, and every figure caption is emitted from a source-owned caption registry. The result is both a map of a genuinely transdisciplinary field and a reproducibility contract: the same version-controlled inputs regenerate the inventories, validation report, analytical figures, manuscript variables, and paper while preserving explicit caveats about registry scope, jurisdictional reach, and the boundary between what the offline gates prove and what only a live source check can confirm.

2 Mapping Entomological Law: Roles, Evidence, and Limits

"Entomological law" — or, from the other direction, "legal entomology" — is not a single discipline but a multi-domain complex in which insects and their biology touch legal norms at dozens of points. A bibliometric analysis of the research literature found that "forensic entomology" and "legal entomology" are used interchangeably and span well over a thousand articles across more than a hundred contributing countries [Magaña et al., 2019]. This reference treats the field in a broad sense: domains in which insects become objects or instruments of legal regulation.

Recent animal-law scholarship now names "insect law" directly, but this reference uses "entomological law" to keep the lens wider than animal-welfare doctrine: the same organism may enter evidence, quarantine, conservation, food, biotechnology, and weapons law before moral status is even at issue [Reddy, 2025].

The deeper connective problem is that courts and agencies repeatedly need insect biology to become what science-and-law scholarship calls a serviceable truth: reliable enough for action, bounded enough to expose uncertainty, and explicit enough to be revisited when the evidence changes [Jasanoff, 2015]. Forensic entomology translates larval development into time; quarantine law translates ecological risk into movement controls; conservation law translates population decline into listing decisions; welfare law translates sentience evidence into moral and statutory thresholds. The same epistemic move recurs across the field.

The second connective problem is classification. Classification systems are not neutral filing cabinets: they allocate visibility, responsibility, and institutional action [Bowker and Star, 1999]. Insects reveal that point with unusual force because they move across ordinary legal boundaries so easily. A fly can be an expert's clock, a statutory animal, a constitutional hook, or a nuisance depending on which classificatory gate opens first. The field therefore also depends on boundary-work: courts and agencies must decide when entomology is sufficiently scientific for evidence, sufficiently uncertain for precaution, sufficiently economic for trade restrictions, or sufficiently moral for welfare concern [Gieryn, 1983, Jasanoff, 2004].

2.1 The role model: what law needs insects to be

The field has no master statute and no single agency, yet it coheres. Its organizing principle is the **legal status the insect occupies in a given dispute**. The same organism is, in turn, a witness, a regulated threat, a protected subject, property, an invention, a defendant, a moral patient, or a weapon — and each role poses a question the legal system was not built to answer. The 8 roles, their domains, and their core questions are listed below and visualized with their registry evidence in the roles-overview figure.

Legal role of the insect	Domain	Core question
Witness / evidence	Forensic entomology	When did death occur, and where?
Regulated threat	Quarantine & invasive-species law	May this organism cross a border?
Protected subject	Conservation / endangered-species law	Does the state owe this species survival?
Property	Common law of wild animals	Who owns this swarm, hive, or specimen?
Invention / product	IP, biotech & food law	Can this insect be patented, engineered, or eaten?
Defendant	Historical animal trials	Can a pest be tried and punished?
Moral patient	Emerging welfare & sentience law	Can an insect be wronged?
Weapon	International humanitarian law & biosecurity	May insects be used in war?

2.2 Why the registry comes first

A typical survey of this material mixes prose summaries of statutes with anecdotes of famous cases; both rot quickly as section numbers are recodified and holdings are distinguished. This reference inverts that pattern. Every legal role, case, statute, taxon, institution, and milestone cited in the prose comes from a Python registry under `src/`, and every count — "18 cases", "38 statutes", "24 taxa" — is a double-brace token resolved at build time by `src.manuscript_variables.generate_variables`. The discipline this enforces is the same reproducibility model the wider template uses: a count that drifts from its registry cannot reach a green PDF without the manuscript-token closure test flipping red first (sec. 12).

Figure 1: The 8 registered legal roles an insect can occupy in this release, with the case, statute, species, and milestone evidence encoded for each. Read as: the field is not one doctrine but a set of recurring legal positions that biology can trigger. Why it matters: the figure sets the manuscript's organizing grammar before the doctrinal sections specialize it. Provenance: `src/roles.py` and `src.metrics.role_coverage_matrix()`. Caveat: counts describe the encoded registries, not the entirety of each legal sub-field.

2.3 A legal history from bee swarms to gene drives

The field is old in more than one way. Its property lineage now reaches back before Rome: the Hittite Laws tariff stolen bees and bee hives, Mishnah Bava Batra treats bees as both beehive property and neighbor-law risk, and Roman law then turns swarms into the classic problem of wildness, sight, and pursuit [Hittite Kingdom, -1650, Mishnah, 200a,b, Justinian, 533a,b]. That lineage also passes through successor-kingdom tariff law: the Salic Law makes stolen bees a named theft subject, and Rothari's Lombard code distinguishes an apiary vessel from bees taken out of a marked tree [Salian Franks, 507, Lombard Kingdom, 643]. Medieval English restatements such as *Fleta* then carry the occupation problem forward, while the evidentiary lineage begins with the 1235 Sickle Murder recounted by Song Ci in *The Washing Away of Wrongs* (1247), where flies settling on an apparently clean blade exposed invisible blood and forced a confession; the claim is bound here both to a Library of Congress record for the Chinese text and to McKnight's scholarly English translation [Fleta, 1290, Ci, 1247, 1981]. Its product-law lineage is old as well: early colonial instruments treated mulberries, silk inputs, silks, and wax as objects of public economic policy rather than merely private agricultural choices [Virginia General Assembly, 1619, Charles II, 1663]. From there the registry traces 34 milestones across 3676 years — bee property, medieval animal trials, colonial silk policy, succession-based time-of-death estimation, twentieth-century conservation and biotechnology statutes, and twenty-first-century sentience and gene-drive debates shown in the timeline figure.

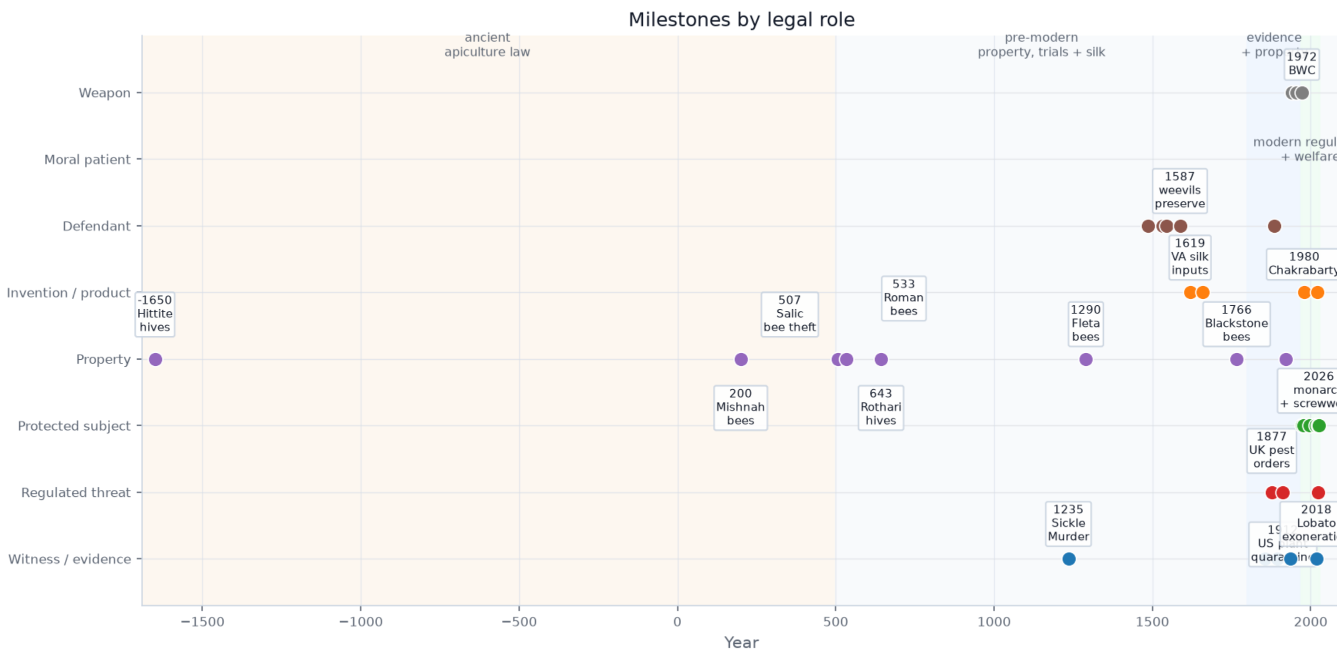

Figure 2: Selected milestones of entomological law across the 3676-year span the registry encodes, from early bee-theft and bee-property rules to the 2026 monarch-listing suit, coloured by legal role. Read as: old evidentiary and property problems keep reappearing inside new biotechnology, welfare, and conservation disputes. Why it matters: the chronology shows recurrence rather than novelty as the field's basic pattern. Provenance: `src/timeline.py`. Caveat: a selected chronology, not a comprehensive history.

2.4 Boundaries of the field map

This reference is a map, not a treatise. The registries are curated to anchor each role with its leading authorities, not to enumerate every decision, instrument, or species; the figures throughout describe the *encoded registries*, and their captions state that scope explicitly. Sections sec. 3 through sec. 10 treat each role in turn; sec. 11 shows how they share legal machinery; and sec. 12 documents the reproducibility contract and its honesty boundary.

3 The Insect as Witness: Forensic Entomology in Court

Forensic entomology is the oldest evidentiary and most legally mature corner of the field: the use of arthropod evidence — chiefly necrophagous blow flies (Calliphoridae) colonizing remains — to estimate the **post-mortem interval (PMI)**, detect corpse movement, prove neglect, and recover toxicology from larvae. Of the 8 legal roles, the witness role is the most case-driven: 5 of the roles carry modern reporter case law, and the witness role alone accounts for 6 of the 18 registered decisions, as shown in the case-by-role and jurisdiction figures.

Figure 3: The 18 registered decisions grouped by the legal role of the insect; 5 of 8 roles carry modern reporter case law, while the defendant, welfare, and weapon roles are history- and statute-driven. Read as: litigation is concentrated where insects enter court as evidence, property, or regulated biological facts. Why it matters: it separates litigated doctrine from roles that are legally important but less case-law dense. Provenance: `src/cases.py` via `src.metrics`. Caveat: only citation-parseable decisions are encoded here.

3.1 From larvae to minimum postmortem interval

Entomologists generally calculate a **minimum PMI** — the time since first colonization — using accumulated degree-days, summing temperature above a developmental threshold and matching it to species-specific growth tables. For older remains they fall back on the succession model of arthropod waves first systematized by Jean-Pierre Mégnin in *La Faune des Cadavres* [Mégnin, 1894]. A sub-branch, entomotoxicology, recovers drugs and metals from maggots when blood and tissue are gone — substances that themselves distort larval growth and so must be accounted for in the estimate. The foundational legal text remains Greenberg and Kunich's *Entomology and the Law*, written expressly to prepare both the entomologist and the trial lawyer for the courtroom [Greenberg and Kunich, 2004].

3.2 Admissibility: Frye, Daubert, and Rule 702

Insect testimony enters US courts through the same gates as all scientific evidence. Under *Frye* the technique must be "generally accepted" in its field [D.C. Circuit, 1923, Cornell Legal Information Institute, 2024b]; under *Daubert* the trial judge is a reliability gatekeeper weighing testability, peer review, error rate, acceptance, and fit [U.S. Supreme Court, 1993]; and *Kumho Tire* extends that inquiry to all expert testimony [U.S. Supreme Court, 1999]. Federal Rule of Evidence 702 codifies the requirement that reliable principles be reliably applied to the facts [Cornell Legal Information Institute, 2024a]. In the reported authorities encoded here, disputes focus less on excluding forensic entomology wholesale than on *application* — wrong geographic dataset, microsite temperature error, species misidentification — and whether reliable principles were reliably applied to the case facts.

Figure 4: The registered case law by issuing court system, spanning the U.S. Supreme Court, federal appellate and state courts, England & Wales, Canada, and the U.S. patent system. Read as: the field is jurisdictionally scattered, so doctrine develops through examples rather than a single appellate line. Why it matters: the field has to be assembled comparatively, not read from one reporter series. Provenance: `src/cases.py`. Caveat: a curated leading-case set, not a census of all decisions.

3.3 Cases where insect evidence mattered

The registry pins the field's evidentiary arc from the 1235 Sickle Murder forward. Bergeret's original *Annales d'hygiène publique et de médecine légale* report is the Western hinge: he treated pupae, larvae, and metamorphosis as medico-legal evidence for a mummified newborn's time of death, making insect development part of a judicial fact pattern rather than a natural-history aside [Bergeret, 1855]. Mégnin then systematized the succession model in a dedicated forensic monograph before modern expert-evidence doctrine existed [Mégnin, 1894]. The modern arc runs through the Buck Ruxton "Jigsaw Murders" that produced the first UK conviction by entomology [Natural History Museum, 2023], the Pennsylvania Supreme Court's affirmation of PMI testimony in *Commonwealth v. Auker* [Supreme Court of Pennsylvania, 1996], the Canadian *R. v. Truscott* reversal decades after conviction [Court of Appeal for Ontario, 2007], and the Kirstin Lobato exoneration built on the telling *absence* of blowfly colonization, which proved death after dark [Innocence Project, 2018]. The field's cautionary tale is *People v. Westerfield*, the celebrated "battle of five entomologists" whose ten-day divergence led the jury to discard the entomology entirely [Supreme Court of California, 2019].

3.4 Standards, certifiers, and error controls

Certification in North America runs through the American Board of Forensic Entomology (ABFE) [American Board of Forensic Entomology, 2024]. The registry encodes 4 forensic-entomology institutions, including the ABFE, the European Association for Forensic Entomology (EAFE), the North American Forensic Entomology Association (NAFEA), and the National Institute of Standards and Technology (NIST) Organization of Scientific Area Committees (OSAC) task group. The 2009 National Academy of Sciences report on forensic science applies with comparatively gentle force here, because entomology's biological grounding is stronger than pattern-matching fields, but its calls for standardized protocols and documented error rates remain only partly met [National Research Council, 2009]. EAFE's peer-reviewed best-practice guideline and field-wide research roadmaps frame the remaining work as a chain-of-custody, sampling, preservation, identification, reporting, and basic-to-applied research problem rather than a single admissibility rule [Amendt et al., 2007, Tomberlin et al., 2011]. Recent global case-report scholarship makes the same point in procedural form: forensic entomology needs comparable reports because litigation turns on whether biological inference can be audited after the fact [Kotzé et al., 2021]. OSAC's proposed standard for collecting and preserving terrestrial entomological evidence makes the same standardization frontier concrete, shifting the problem from whether protocols can be specified to how consistently investigators apply them [Organization of Scientific Area Committees for Forensic Science, 2025].

4 The Insect as Threat: Quarantine, Invasion, and Vector Control

Where forensic law asks insects to *speak*, regulatory law tries to *stop them from moving*. The architecture is dense and federal. In the registry the threat role is statute-driven rather than case-driven: it is anchored by 14 instruments spanning the quarantine and public-health categories, summarized in the statutes-by-category figure.

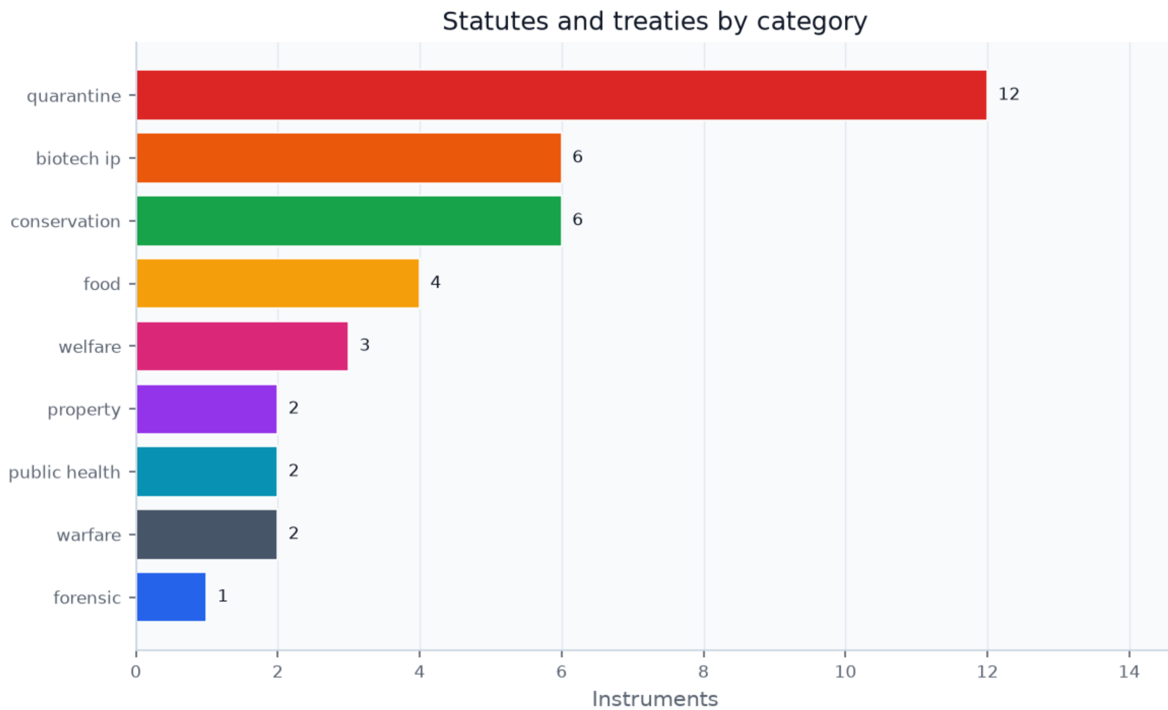

Figure 5: The 38 registered instruments grouped across 9 structural categories — forensic, quarantine, conservation, property, biotech/IP, food, welfare, public-health, and warfare. Read as: regulation follows the legal problem insects pose, not insect taxonomy itself. Why it matters: it shows why the same organism can move between food, biosecurity, conservation, and weapons law. Provenance: `src/statutes.py` via `src.metrics`. Caveat: a representative backbone, not every instrument in each category.

4.1 US quarantine authority and plant-pest movement

The modern statute sits on a pre-1950 statutory architecture rather than a blank slate. Britain's Destructive Insects Act made Colorado beetle an object of border prohibition, crop destruction, entry on land, official records, compensation, and penalties; India's Destructive Insects and Pests Act then cast insects, fungi, and other crop pests as import and interprovincial-movement risks [United Kingdom Parliament, 1877, India Legislative Department, 1914]. In the United States, the Federal Insecticide Act treated insect-killing chemistry as a misbranding and adulteration problem in commerce, while the Plant Quarantine Act gave USDA authority over import exclusion, interstate quarantine, inspection, disinfection, certification, and a Federal Horticultural Board that included the Bureau of Entomology [U.S. Congress, 1910, 1912]. The point is analytical: before modern environmental law, insect threat was already split among border control, crop police powers, chemical-product integrity, and administrative entomology.

The master contemporary statute is the Plant Protection Act, cited here at 7 U.S.C. §§ 7701–7786, which empowers the U.S. Department of Agriculture's Animal and Plant Health Inspection Service (APHIS) to prohibit movement, declare quarantines, require permits, and seize or destroy infested articles [U.S. Congress, 2000a,b]. Moving a live insect interstate or importing one requires an APHIS permit and an approved containment facility. The rogues' gallery of regulated pests is encoded in the species registry: the spotted lanternfly under active quarantine [USDA APHIS, 2026], the Asian longhorned beetle under eradication, the emerald ash borer whose *federal* quarantine was lifted in 2021, and the northern giant hornet — the "murder hornet" — declared eradicated from the United States in 2024, the first *Vespa* eradication in North America [The Guardian, 2024].

4.2 International, IPPC, and EU risk layers

Above the US sits the International Plant Protection Convention, whose International Standards for Phytosanitary Measures (ISPMs) serve as the benchmark under the World Trade Organization (WTO) Agreement on the Application of Sanitary and Phytosanitary Measures (SPS Agreement) for judging whether a quarantine is science-based or a disguised trade barrier [IPPC, 2026a]. IPPC's pest-status standard makes pest records, status categories, and uncertainty part of the legal infrastructure for deciding whether a pest is present in an area [IPPC, 2026b]. The European Union (EU) runs its own priority-pest list and plant-passport system under the Plant Health Law [European Union, 2016]. The EU also treats invasive alien species as a separate Union-wide risk category: Regulation (EU) 1143/2014 creates a Union list whose species face restrictions on keeping, importing, selling, breeding, growing, and release, and the European Commission says the fourth update of that list entered into force on 7 August 2025 [European Union, 2014, European Commission, 2026a].

That architecture makes invasive-insect law a risk-allocation system, not merely a list of pests. The legal decision is whether uncertain ecological evidence justifies stopping trade, seizing property, requiring treatment, or spending public money on surveillance and eradication. Bioeconomic scholarship on invasive species frames that choice as an institutional problem of pathways, probabilities, expected harms, and management cost under uncertainty [Lodge et al., 2016]. Insects make the problem unusually sharp because the same shipment, nursery stock, package, or animal wound can be legally ordinary until an expert identifies a life stage, pathway, or reproductive risk that moves it into quarantine law.

4.3 Liability gaps when pest movement is indirect

The Lacey Act's injurious-species provision at 18 U.S.C. § 42 is a strict-liability criminal statute, but it conspicuously **does not list insects**, deferring to the Plant Protection Act for plant pests and leaving non-plant-pest insects in a regulatory gap [U.S. Congress, 1900, Congressional Research Service, 2018]. Recent enforcement marks the limits of secondary liability: in *Amazon Services LLC v. USDA* the D.C. Circuit held that aiding or inducing a regulated-pest movement requires conscious and culpable participation, not mere fulfilment services [D.C. Circuit, 2024]. And in a Minnesota tree-infestation suit the claim failed for lack of an entomological expert — proof that forensic and regulatory entomology converge, because invasive-pest causation cannot be shown without expert insect testimony [Minnesota Lawyer, 2024].

4.4 Vectors as public-health infrastructure

The same regulatory logic extends to insects as disease vectors. California has operated mosquito-abatement and vector-control districts since 1915 under a statute declaring organized public programs the best protection against vector-borne disease [California Legislature, 1915], and federal mosquito-control funding has been advanced as national public-health law [ASTHO, 2020]. International animal-health law adds a standards layer: WOAH's Terrestrial Code includes a chapter on surveillance for arthropod vectors of animal diseases, showing how vector law crosses animal health, human health, and trade [World Organisation for Animal Health, 2026]. This vector-control strand is where the threat role brushes against the weapon role of sec. 10.

5 The Insect as Protected Subject: Conservation and Recovery Law

The mirror image of quarantine law: instead of exterminating insects, the state guarantees their survival. The species registry encodes 6 taxa in the protected role — from the Schaus swallowtail, among the first insects ever listed, to the rusty patched bumble bee, the monarch, and the American burying beetle, summarized in the species-by-role figure.

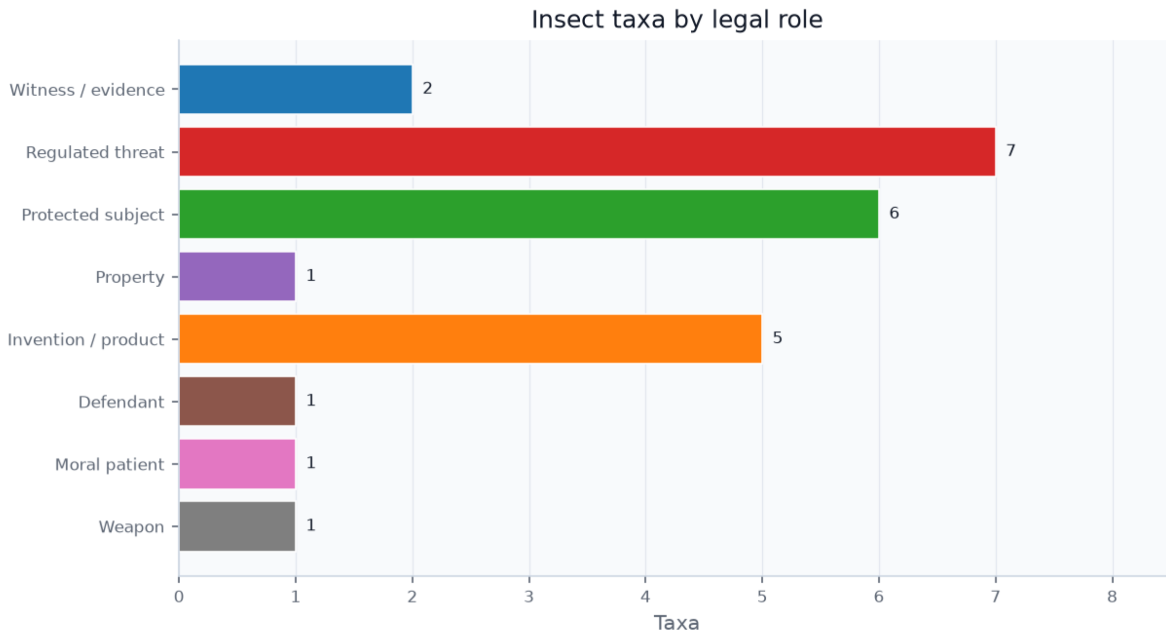

Figure 6: The 24 registered taxa grouped by the legal role they illustrate, from forensic blow flies to quarantined pests, listed species, owned honey bees, engineered and edible insects, the trial weevils, and a weaponized flea. Read as: the same biological class can become evidence, asset, enemy, product, or rights candidate depending on legal context. Why it matters: legal role, not species identity alone, determines which institution acts. Provenance: `src/species.py`. Caveat: an illustrative set chosen to anchor each role, not a taxonomic survey.

5.1 The ESA definition that reaches arthropods

The federal Endangered Species Act (ESA), cited here at 16 U.S.C. § 1531, defines "fish or wildlife" to expressly include any "arthropod or other invertebrate," putting non-pest insects within the statute's eligibility frame [U.S. Congress, 1973a,b]. Conservation-law scholarship has treated that textual inclusion as more than a curiosity: it is the doctrinal opening through which ecologically central but politically obscure organisms can become federal legal subjects [Lugo, 2006]. Recent conservation science now reviews invertebrate listing history and threats as an ESA problem in its own right, reinforcing that insects are not merely edge cases inside vertebrate-centered conservation law [Shirey et al., 2025]. Among the first insects listed was the Schaus swallowtail in 1976 (with the Bahama swallowtail); the Delhi Sands flower-loving fly, listed in 1993, became the first and only fly — and the unlikely protagonist of a constitutional landmark.

5.2 Commerce Clause protection for tiny species

In *National Association of Home Builders v. Babbitt*, cited at 130 F.3d 1041, developers argued Congress lacked Commerce Clause power over a fly that lives entirely within California; a divided panel upheld the protection, and legal scholarship quickly recognized the Delhi Sands flower-loving fly as a test of whether tiny, local, economically inconvenient species could carry national ecological value [D.C. Circuit, 1997, Nagle, 1998]. The "take" prohibition reaches habitat modification under *Babbitt v. Sweet Home*, so insect habitat enjoys the same protection as the insects themselves [U.S. Supreme Court, 1995]. State law is widening the institutional map as well: Colorado's invertebrate conservation law adds rare plants and invertebrates to its nongame conservation statute and authorizes voluntary programs to conserve, protect, and perpetuate invertebrates [Colorado General Assembly, 2024].

5.3 State law and the bumblebee-as-fish problem

California produced the field's most creative ruling. In *Almond Alliance of California v. Fish & Game Commission*, cited at 79 Cal.App.5th 337, the Court of Appeal held that **bumblebees are "fish"** under Fish & Game Code § 45, because that section's definition of "fish" includes "invertebrate" — allowing four *Bombus* species to be listed under the California ESA [California Court of Appeal, Third District, 2022, California Legislature, 1957, Xerces Society, 2022]. It is the single most-cited "insect is a fish" holding in the field and links conservation law directly to the definitional question that recurs throughout (sec. 11). The monarch butterfly was proposed for threatened listing in 2024; as of June 29, 2026, the U.S. Fish and Wildlife Service (FWS) still described the species as proposed and said protections would not apply until a final rule became effective [U.S. Fish and Wildlife Service, 2024, 2026].

5.4 Trade, pollinators, and insect decline

The Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) supplies the trade-law counterpart: its official checklist identifies *Ornithoptera alexandrae* — Queen Alexandra's birdwing — as an insect with current listing I, and domestic implementing legislation then makes covered trade legal, sustainable, and traceable [CITES Parties, 1973, CITES Secretariat and UNEP-WCMC, 2026]. The global biodiversity layer broadens the frame: the Kunming-Montreal Global Biodiversity Framework gives insect conservation a treaty-scale target architecture, while insect-focused conservation scholarship warns that existing biodiversity indicators may fail to show whether policy is actually recovering insect populations unless insect-focused indicators are built [Convention on Biological Diversity, 2022, Bladon et al., 2026]. EU conservation law is now more explicit about pollinators as a protected infrastructure: the Nature Restoration Regulation sits alongside the EU Pollinators Initiative, and the European Commission says that EU policy commits to reversing wild-pollinator decline by 2030 while reporting that 1 in 3 bee, butterfly, and hoverfly species is proven to be in decline and 1 in 10 bee and butterfly species is threatened with extinction [European Union, 2024, European Commission, 2026c]. Driving all of this is the science of the "insect apocalypse": a landmark study found a decline of more than 75 percent — a 76 percent seasonal and 82 percent mid-summer fall — in flying-insect biomass over 27 years of monitoring in protected areas [Hallmann et al., 2017], a finding amplified by global reviews and scientists' warnings about entomofauna decline and its interacting pressures [Sánchez-Bayo and Wyckhuys, 2019, Wagner et al., 2021, Cardoso et al., 2020]. The legal-protection gap is now measurable as well as rhetorical: a 2026 PNAS study reports that the conservation status of 88.5 percent of described North American insect and arachnid species is unknown, while 94.7 percent of U.S. insects and arachnids at-risk throughout their range are not protected by any state or federal law [Entomological Society of America, 2021, Walsh and Figueroa, 2026].

The legal difficulty is that insect value is often infrastructural rather than charismatic. Pollination, pest suppression, nutrient cycling, waste processing, and food-web support are ecological services before they are individualized legal interests, which means law must translate diffuse background work into administrable species, habitat, trade, and take decisions [Losey and Vaughan, 2006]. That translation explains why this role touches both property and threat: the same insect can be valuable enough to protect in one setting and disruptive enough to suppress in another.

6 The Insect as Property: Bees, Specimens, and Qualified Possession

When an insect is neither evidence, threat, nor protected subject, it may simply be *owned*. No insect has generated more property law than the honeybee, and the registry pins 5 decisions to this role.

6.1 Capture, hiving, pursuit, and possession

The oldest directly verified property layer is not Roman. The Hittite Laws treat stolen bees in a swarm and stolen bee hives as tariffed wrongs, while Hittite apiculture scholarship shows that honey and bees were economic and ritual resources inside a wider cuneiform record [Hittite Kingdom, -1650, Demirel, 2022]. Mishnah Bava Batra adds a legal-contract layer: a sold beehive carries the bees in it, and beehive produce is handled through swarms and honeycombs [Mishnah, 200b]. Roman law then classed bees as *ferae naturae* — wild by nature, owned only through capture and lost when they abandon the *animus revertendi*, the intention to return. The Institutes say a swarm on another's tree is not owned until hived, while the Digest adds the pursuit rule that a swarm leaving a hive remains the keeper's only while it is visible and not difficult to follow [Justinian, 533a,b]. Early medieval law did not merely repeat that Roman template. The Salic Law gives stolen bees their own theft title, while Rothari's Lombard code separates theft from an apiary vessel from taking bees out of another person's marked tree; those clauses make enclosure, marking, and woodland claim legible before the later common-law vocabulary arrives [Salian Franks, 507, Lombard Kingdom, 643]. Irish *Bechbretha* treats hives, swarms, bee trespass penalties, and distraint of bees as legal subjects; Bracton repeats the hiving and pursuit rules in English legal vocabulary; *Fleta* gives the same occupation rule in medieval English legal Latin, using bees to show that enclosure and practicable pursuit matter more than mere presence on the owner's land; and the Welsh laws give both valuation rules for bee stocks and swarms and recovery rules for bees that enter another person's skep [Charles-Edwards and Kelly, 1983, Bracton, 1250, Fleta, 1290, Commissioners on the Public Records of the Kingdom, 1841]. Recent scholarship on early medieval apiculture captures the broader shift: successor-kingdom legislation often moved bees from pure wildness toward beekeeper property, while comparative Indo-European scholarship treats Hittite, Irish, and later bee-law parallels as suggestive but not proof of a single inherited rule [Martinez Jimenez, 2022, Joseph, 2018]. Blackstone then carried the Roman structure into common-law commentary, describing hived and reclaimed bees as qualified property under natural and civil law [Blackstone, 1766]. The common law also carried the broader wild-animal doctrine forward through the Case of Swans [cas, 1592]. The leading modern English authority, *Kearry v. Pattinson* at [1939] 1 KB 471, holds that there is **no right to pursue a swarm onto a neighbour's land** [King's Bench, 1939]; the leading American authority, *Goff v. Kilts* at 15 Wend. 550, recognizes a **qualified property** in bees so long as the owner keeps them in sight and pursues them [New York Supreme Court, 1836].

That doctrine is not just a bee-specific oddity. Property theory has long treated possession as a communicative act: the would-be owner must give the world a recognizable signal that a thing has been appropriated [Rose, 1985]. Bees strain that theory because the signal is biological and relational rather than purely physical. A hive, a marked box, visible pursuit, disease records, and apiary registration all become ways of making mobile insects legible as property.

The same source line also reaches tithe and fiscal argument. Selden's *Historie of Tithes* and Elderfield's later defense of tithes both preserve the bee clause as part of a wider early-modern debate over whether profit from bees belongs with woods, meadows, waters, mills, fisheries, gardens, and trade [Selden, 1618, Elderfield, 1650]. That matters analytically because it moves bees from possession alone into accounting: law is not only asking who captured the swarm, but which institution can claim a share of the bee-derived yield.

6.2 Neighbor disputes, regulation, and liability

The neighbor-injury pattern is much older than modern apiary statutes. Mishnah Bava Batra's mustard-and-bees rule places bees inside neighbor-law distance doctrine, and Bava Batra's later discussion makes the conflict reciprocal: one side can complain about mustard harming bees, while the other can complain about bees eating mustard plants [Mishnah, 200a, Babylonian Talmud, 500]. Quintilian's legal declamation imagines a poor beekeeper whose bees die after a rich neighbor poisons flowers, framing the loss as wrongful damage rather than a mere natural misfortune [Quintilian, 100]. Modern law translates that same problem into regulation and tort. The federal Honeybee Act restricts bee importation, and states layer apiary registration, disease protocols, and Africanized-bee controls on top [U.S. Congress, 1922]. Keeping bees near neighbours carries tort exposure: in *Ferreira v. D'Asaro* a Florida court addressed liability for sting injuries from hives maintained close to a residential property line [Florida District Court of Appeal, 1963]. The same property/tort logic reaches the structures insects damage — a pest-control firm faced professional liability for a negligent wood-destroying-insect inspection in *Horsch v. Terminex* [Kansas Court of Appeals, 1993]. Bees are now serious crime targets as well: within California's almond-pollination economy, large-scale hive thefts have been prosecuted as agricultural or livestock theft.

6.3 Owned honeybees and protected wild bees

The honeybee one can own and the rusty patched bumble bee the state now protects (sec. 5) sit on opposite ends of the same *ferae naturae* doctrine — a mirror symmetry taken up directly in sec. 11.

7 The Insect as Invention and Product: Biotech, Food, Feed, and Silk

The newest frontier: insects deliberately engineered, patented, farmed, and eaten. The registry pins 3 foundational decisions and 5 taxa to this role, from the Oxitec mosquito to the house cricket, the yellow mealworm, the black soldier fly, and the silkworm.

7.1 Patents for engineered living things

Since *Diamond v. Chakrabarty*, cited at 447 U.S. 303, living organisms are patentable subject matter — "anything under the sun that is made by man" [U.S. Supreme Court, 1980]. *Ex parte Allen* extended this to multicellular animals [Board of Patent Appeals, 1987], and the Harvard OncoMouse confirmed it for whole engineered animals [Leder and Stewart, 1988]. Insects sit squarely within this regime — genetically modified and sterile-insect lines, hundreds of *Bacillus thuringiensis* patents, silk inventions, and a fast-growing insect-biomimicry sector — bounded only by the statutory bar on patenting humans.

7.2 Engineered mosquitoes and gene drives

The 1986 Coordinated Framework splits jurisdiction over genetically modified insects among the Environmental Protection Agency (EPA), APHIS, and the Food and Drug Administration (FDA) [Office of Science and Technology Policy, 1986]. The flagship case is Oxitec's engineered *Aedes aegypti*, released in the Florida Keys under an EPA experimental-use permit [U.S. Environmental Protection Agency, 2020]. WHO guidance now supplies the public-health testing baseline for genetically modified mosquitoes, emphasizing safe, ethical, and rigorous staged evaluation before deployment [World Health Organization, 2021]. EPA's current pesticide-law posture is correspondingly procedural and risk-pathway specific: its FIFRA Scientific Advisory Panel materials ask how to determine the absence of novel proteins in the saliva of genetically engineered female mosquitoes, while a May 27, 2026 EPA fact check stated that the prior experimental-use permit had expired and that no genetically engineered mosquito releases were then authorized in the United States [U.S. Environmental Protection Agency, 2025, 2026]. **Gene drives** — constructs that force inheritance through wild populations — are governed by general biosafety and biotechnology instruments rather than a deployment-specific insect regime, a governance mismatch emphasized by early regulatory proposals, consensus reports, and responsible-innovation scholarship [Oye et al., 2014, Convention on Biological Diversity, 2000, National Academies of Sciences, Engineering, and Medicine, 2016, Fisher, 2018]. A later implementation review for gene-drive modified mosquitoes sharpens the institutional point: product definition, transboundary movement, market entry, and the role of national, regional, and multinational authorities are design questions, not late-stage paperwork [James et al., 2023]. The Sterile Insect Technique underpins the screwworm-eradication program, which re-entered the U.S. legal and veterinary agenda when New World screwworm was confirmed in a calf in Zavala County, Texas, on June 3, 2026; the Texas Animal Health Commission later described an established infested zone, quarantine limits on movement of warm-blooded animals, and joint federal-state surveillance and response work [Texas Animal Health Commission, 2026].

7.3 Insect food and feed as regulated markets

The EU leads via the Novel Food Regulation, cited at Regulation (EU) 2015/2283, which has authorized several insect products — house cricket, yellow mealworm, migratory locust, and lesser mealworm among them — each through an individual implementing regulation and European Food Safety Authority (EFSA) opinion [European Union, 2015, European Commission, 2026b]. Comparative food-law scholarship treats the EU's authorization model as unusually explicit relative to many national systems [Lähteenmäki-Uutela et al., 2021]. The United Nations food-and-feed literature supplies the policy backdrop: insects are framed simultaneously as food security infrastructure, waste-conversion technology, and farmed animals [van Huis et al., 2013]. The US does not have a single insect-food pathway comparable to the EU's novel-food authorizations: edible insects sit awkwardly between the adulteration and "filth" provisions of the Food, Drug, and Cosmetic Act and the Generally Recognized as Safe (GRAS) framework [U.S. Congress, 1938, U.S. Food and Drug Administration, 2018]. But US state law is beginning to name insect protein as a product category: Utah's H.B. 138 requires labeling for foods containing plant or insect-based meat substitutes, making insect substitutes legible as alternative-protein products rather than mere contaminants [Utah Legislature, 2025]. On the feed side, the EU has progressively relaxed its post-bovine spongiform encephalopathy (BSE) feed ban to admit processed insect protein, beginning with aquaculture [European Union, 2017]. A recent EU legal-barriers study adds a live-feed boundary: SCoPAFF clarified that live insects may be used as feed in the EU except for ruminants [Ziety et al., 2026].

7.4 Genetic resources, silk, and colonial development

Bioprospecting insect venom peptides, antimicrobials, and silk proteins implicates the Nagoya Protocol's sovereignty and benefit-sharing duties [Convention on Biological Diversity, 2010]. Sericulture is a historically deep regulatory domain in its own right: Virginia's first assembly already made mulberry planting and silk-flax work a legal development duty, Hartlib's Virginian silk-worm pamphlet framed silkworm rearing and mulberry planting as public industry, Virginia's later mulberry-tree act continued that statutory infrastructure, and the Carolina charter treated silks and wax as duty-favored colonial commodities [Virginia General Assembly, 1619, Hartlib, 1655, Virginia Grand Assembly, 1658, Charles II, 1663]. Georgia Trustee records show the same administrative impulse in practice, tracking public mulberry nurseries and settler progress in silk culture, while Atlantic-world sericulture scholarship explains why imperial administrators kept trying to make silkworm cultivation a development program [Egmont, 1962, Marsh, 2020a,b]. India's Central Silk Board Act continues that pattern by making quality-controlled silkworm biology a matter of statutory public interest [Parliament of India, 1948]. This domain loops back to conservation (the biotechnology value of species was a Commerce Clause hook in sec. 5) and forward to welfare (sec. 9), since industrial insect farming raises animals at a scale no prior welfare regime contemplated.

8 The Insect as Defendant: Animal Trials and Legal Ritual

Across late medieval and early modern Europe, courts literally **prosecuted insects**. This is the field's most astonishing chapter, and its most history-driven: the registry encodes no modern reporter case for the defendant role, but 5 timeline milestones and the trial weevils themselves carry it. The pre-1900 source base is stronger than a single retrospective curiosity. Ménabréa's 1846 Savoy academy study organizes the material around proceedings against insects, civil and criminal animal proceedings, pleadings for and against the insects, prosecutorial conclusions, and an ecclesiastical sentence; Evans's 1884 Atlantic article then makes that procedural archive available in English before his later book becomes the canonical synthesis [Ménabréa, 1846, Evans, 1884, 1906, Beirne, 1994].

8.1 Chassenée and the prosecuted weevils

The jurist Bartholomew Chassenée built his career defending vermin and wrote the first legal treatise on insect prosecution, arguing that animals were "lay persons" entitled to counsel before ecclesiastical courts. Evans's 1884 account is useful because it treats the Chassenée episode as law rather than folklore: summons, notice, default, counsel, jurisdiction, judge, and forum all matter before any malediction can issue [Evans, 1884]. Ménabréa supplies the complementary French documentary frame, including the St-Julien materials and a preserved "plea for the insects" that makes the insects procedurally visible even though they remain nonrational pests [Ménabréa, 1846]. The vine-weevils of St-Julien were first treated as a divine scourge and later met with a proposed preserve for the insects; earlier proceedings against the slugs of Autun round out the registry's defendant milestones [Evans, 1906].

8.2 What animal trials reveal about modern law

These trials are not merely curiosities. They separate **procedure** from **personhood** with unusual clarity: the court can stage notice, representation, and sentence around an insect population without deciding that insects possess moral agency or ordinary legal capacity. That is why the defendant role matters to modern law. The 1587 offer of a land preserve for the weevils foreshadows modern **critical-habitat designation** (sec. 5), and the medieval debate over whether an animal can be a defendant prefigures the modern debate over whether it can be a *rights-holder* (sec. 9). Entomological law repeatedly uses legal form to manage biological agency: the insect can be summoned, protected, patented, quarantined, or excluded before the legal system has a stable theory of what kind of subject it is. The defendant role is thus the historical hinge of the field's two most forward-looking questions, a connection drawn out in sec. 11.

9 The Insect as Moral Patient: Sentience, Welfare, and Legal Standing

The field's living edge. The question is no longer whether an insect can testify or be owned, but whether it can be *wronged*. The registry pins 1 taxon to this role as an exemplar — the fruit fly — and grounds the role in emerging legislation rather than reporter case law.

9.1 Sentience evidence and statutory expansion

A London School of Economics (LSE) review built a multi-criterion sentience framework from a large body of studies [Birch et al., 2021]. The UK Animal Welfare (Sentience) Act 2022, cited at Animal Welfare (Sentience) Act 2022, acted on it: its interpretation section defines "animal" to include any vertebrate other than humans, any cephalopod mollusc, and any decapod crustacean, and reserves a regulatory power to extend coverage to other invertebrates [UK Parliament, 2022]. The Animal Sentience Committee now makes the insect question institutionally visible: in January 2026 it said discussions around animal sentience were gathering momentum for some insects and arachnids, and recommended clear processes for reviewing sentience evidence so legislation can be updated in time [Animal Sentience Committee, 2026]. The US Animal Welfare Act draws the opposite boundary from within federal animal-welfare law: USDA describes the statutory definition as centered on named taxa and other warmblooded animals, leaving insects outside that baseline [USDA National Agricultural Library, 2026]. Strikingly, the comparative-evidence literature finds that adult flies and cockroaches satisfy a majority of the established sentience criteria — scoring higher than some of the very decapods the Act already covers — and has begun to ask whether routine insect research should itself adopt welfare review [Gibbons et al., 2022, Crump et al., 2023]. Recent ethical scholarship pushes the same debate beyond proof of pain alone, arguing that precaution and intrinsic value can matter when invertebrate sentience remains hard to demonstrate conclusively [de Souza Valente, 2025].

9.2 Exclusion, scale, and the farmed-insect problem

EU law shows the same exclusion. Directive 2010/63/EU, governing animals in scientific procedures, systematically leaves out insects [European Union, 2010], and the Treaty on the Functioning of the European Union (TFEU)'s recognition of animals as "sentient beings" is not self-executing and in practice protects almost exclusively vertebrates [European Union, 2007]. The stakes are scale: Rethink Priorities estimates that roughly 1 trillion to 1.2 trillion insects are raised on farms annually for food and feed, prompting farmed-insect welfare proposals that must move beyond sentience alone to slaughter, stocking density, handling, and environmental conditions [Rethink Priorities, 2026, van Huis, 2021, Barrett and Fischer, 2023]. Animal-law critique of insect-based agriculture adds a second warning: sustainability framing can obscure industrial scale, feed-chain substitution, and welfare costs rather than resolving them [Reddy, 2024]. The legal imagination is broader still: the path from property to personhood runs through proposals for living-property status, rights for natural objects, legal personhood at the boundaries of biological and artificial agency, and evolutionarily inclusive ethics [Favre, 2010, Stone, 1972, Rowe, 2025, Mikhalevich and Powell, 2020]. This domain is the philosophical counterweight to the commercialization of sec. 7 — the same industrial farming that food and feed law encourages is what welfare law now scrutinizes.

10 The Insect as Weapon: Vectors, Crops, and International Humanitarian Law

Finally, insects as instruments of war, biosecurity, and international humanitarian law (IHL). The registry carries this role through 3 historical milestones and 1 taxon — the plague-bearing oriental rat flea — rather than through litigation.

10.1 Entomological warfare before biotechnology

Japan's Unit 731 conducted plague-flea air-drops; the Ningbo attack of 1940 alone caused major civilian casualties, with broader operations linked to large-scale civilian deaths. Cold War programs followed, including the US Operation Big Itch, which tested uninfected-flea dispersal.

10.2 Treaty limits and dual-use insects

The Biological Weapons Convention (BWC), cited at Biological Weapons Convention (1972), treats insect vectors as covered means of delivery under its General Purpose Criterion, though ambiguity persists around uninfected insects used in crop warfare [States Parties to the Biological Weapons Convention, 1972]. That ambiguity erupted publicly over the Defense Advanced Research Projects Agency (DARPA)'s "Insect Allies" program, which proposed using insects to deliver protective genes to crops; a 2018 critique in *Science* argued the dual-use technology risked violating the Convention, a charge the program disputed [Reeves et al., 2018]. Domestically, the Agricultural Bioterrorism Protection Act classifies certain plant pests as select agents subject to stringent possession and transfer rules [U.S. Congress, 2002]. Insect-warfare law thus connects to the gene-drive debates of sec. 7 — the same biotechnology read alternately as agriculture and as weapon, and loosely overseen by the Cartagena Protocol [Convention on Biological Diversity, 2000].

11 Interconnections: How Insects Move Between Legal Roles

The 8 roles are not silos. They share recurring legal machinery, and the registry encodes 5 themes that knit them together in the interconnections figure. The role-coverage matrix makes the field's structure visible at a glance: some roles are case-driven, some statute-driven, and some — the defendant and the weapon — almost entirely history-driven.

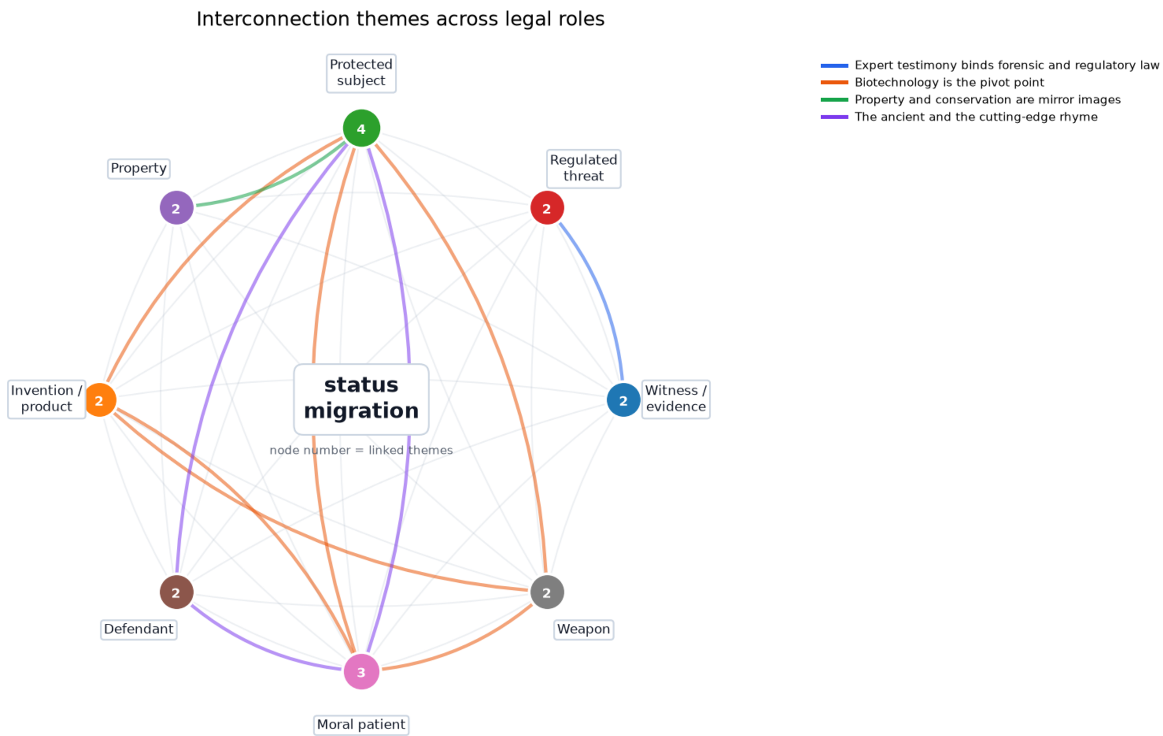

Figure 7: Network of the 5 recurring themes that link the legal roles — the definitional problem, the expert-testimony bridge, the biotechnology pivot, the property/conservation mirror, and the ancient/modern rhyme. Edge bundles share a theme colour. Read as: category changes, not species names, are what move a dispute from one legal regime into another. Provenance: `src/interconnections.py`. Why it matters: the synthesis depends on transfers among categories rather than parallel lists of topics. Caveat: the graph encodes declared thematic links, not statistical association.

The themes, and the roles each links, are catalogued below.

Theme	Roles linked	Description
The definitional problem	Witness / evidence, Regulated threat, Protected subject, Property, Invention / product, Defendant, Moral patient, Weapon	Entomological law is, at root, a series of fights over what category a bug belongs to: is a bumblebee a 'fish', a screwworm a 'plant pest', a fly 'wildlife', an insect 'made by man', a cricket 'sentient'?
Expert testimony binds forensic and regulatory law	Witness / evidence, Regulated threat	Proving invasive-pest causation requires the same entomological expertise as proving time of death — a tree-infestation suit failed for lack of an insect expert.
Biotechnology is the pivot point	Invention / product, Protected subject, Weapon, Moral patient	GM and gene-drive insects are simultaneously regulated products, conservation tools or threats, potential weapons, and moral patients in farmed-welfare debates.

Theme	Roles linked	Description
Property and conservation are mirror images	Property, Protected subject	Roman and common-law bee possession rules sit opposite modern bumblebee protection under the same ferae naturae doctrine.
The ancient and the cutting-edge rhyme	Defendant, Protected subject, Moral patient	Roman bee pursuit, Irish bee trespass, and the 1587 weevil preserve foreshadow modern fights over habitat, rights, and legally actionable insect status.

- **The definitional problem** runs through everything: is a bumblebee a "fish" (sec. 5), a screwworm a "plant pest" (sec. 4), a fly "wildlife" (sec. 5), an insect "made by man" and patentable (sec. 7), or "sentient" (sec. 9)? Entomological law is, at root, a series of fights over what category a bug belongs to.
- **Expert testimony binds forensic and regulatory law**: proving invasive-pest causation requires the same entomological expertise as proving time of death (sec. 3, sec. 4).
- **Biotechnology is the pivot point**: engineered and gene-drive insects are simultaneously regulated products, conservation tools or threats, potential weapons, and moral patients (sec. 7, sec. 5, sec. 10, sec. 9).
- **Property and conservation are mirror images**: the honeybee one can own (sec. 6) and the bumblebee the state protects (sec. 5) sit on opposite ends of the same *ferae naturae* doctrine, a line running from Hittite, rabbinic, Salic, and Lombard bee clauses through Justinian, Fleta, and Blackstone to modern listing law [Hittite Kingdom, -1650, Mishnah, 200b, Salian Franks, 507, Lombard Kingdom, 643, Justinian, 533a, Fleta, 1290, Blackstone, 1766].
- **The ancient and the cutting-edge rhyme**: Hittite hive theft, rabbinic bee nuisance, Roman bee pursuit, Irish bee trespass, Quintilian's poisoned-flower bee claim, and the St-Julien weevil proceedings all foreshadow modern fights over habitat, rights, and legally actionable insect status (sec. 8, sec. 5, sec. 9) [Hittite Kingdom, -1650, Mishnah, 200a, Justinian, 533b, Charles-Edwards and Kelly, 1983, Quintilian, 100, Ménabréa, 1846, Evans, 1884].

The connective tissue is not metaphorical. It is a recurring translation from biological fact to legal status. A larva becomes a clock; a mosquito becomes either public-health infrastructure or a releaseable regulated article; a fly becomes wildlife; a cricket becomes food, feed, and possibly a welfare subject. That is why the same evidentiary problem appears in unrelated doctrinal settings: law needs science to produce action-ready, reviewable claims without pretending uncertainty has disappeared [Jasanoff, 2015]. The Delhi Sands flower-loving fly made that problem constitutional, the EU Union list makes it administrative, the gene-drive debate makes it anticipatory, and the farmed-insect welfare debate makes it industrial [Nagle, 1998, European Commission, 2026a, Oye et al., 2014, Barrett and Fischer, 2023].

The deeper arc is status migration. Wild animals begin as unowned things, become qualified property when captured, become protected subjects when scarcity matters, become products when engineered or eaten, and become rights candidates when sentience or ecological standing enters the frame. The pre-modern sources show that this migration is not new: bees were already stolen swarms, hive contents, neighbor-law hazards, marked-tree resources, tithable yields, trespassers, and objects of neighborly remedy before modern conservation and biotech vocabularies existed, while colonial silk and wax instruments show insect-derived commodities becoming public economic policy before modern biotechnology law had a name [Hittite Kingdom, -1650, Mishnah, 200b,a, Salian Franks, 507, Lombard Kingdom, 643, Selden, 1618, Elderfield, 1650, Virginia General Assembly, 1619, Charles II, 1663]. The animal-trial sources add a separate lesson: legal procedure can organize a conflict around insects even when doctrine refuses to treat insects as morally accountable persons [Ménabréa, 1846, Evans, 1884]. Legal scholarship on living property and standing for natural objects helps explain why insects are such good stress tests: they sit at the edge of every familiar boundary while remaining biologically central to ecosystems, agriculture, evidence, and public health [Favre, 2010, Stone, 1972, Cardoso et al., 2020, van Huis et al., 2013].

The hard cases also show why entomological law cannot be reduced to "nature law" or "animal law." A quarantine officer, a forensic expert, a conservation biologist, a food regulator, and a welfare theorist are often looking at the same biological fact but asking different institutional questions: admissibility, movement risk, extinction risk, market authorization, or moral considerability. Scholarship on invasive-species risk analysis, insect conservation, comparative food-and-feed regulation, and invertebrate ethics supplies the missing bridge: each field has built a translation apparatus for moving from uncertain insect science to a legal decision [Lodge et al., 2016, Lugo, 2006, Lähteenmäki-Uutela et al., 2021, Mikhalevich and Powell, 2020].

The newest sources sharpen that bridge. The Global Biodiversity Framework turns insect recovery into an indicator problem, because global targets cannot protect insects if monitoring systems do not detect insect-specific decline or recovery [Convention on Biological Diversity, 2022, Bladon et al., 2026]. Utah's alternative-protein labeling statute shows the same classification work in market form: a cricket or mealworm ingredient must be named differently from conventional meat

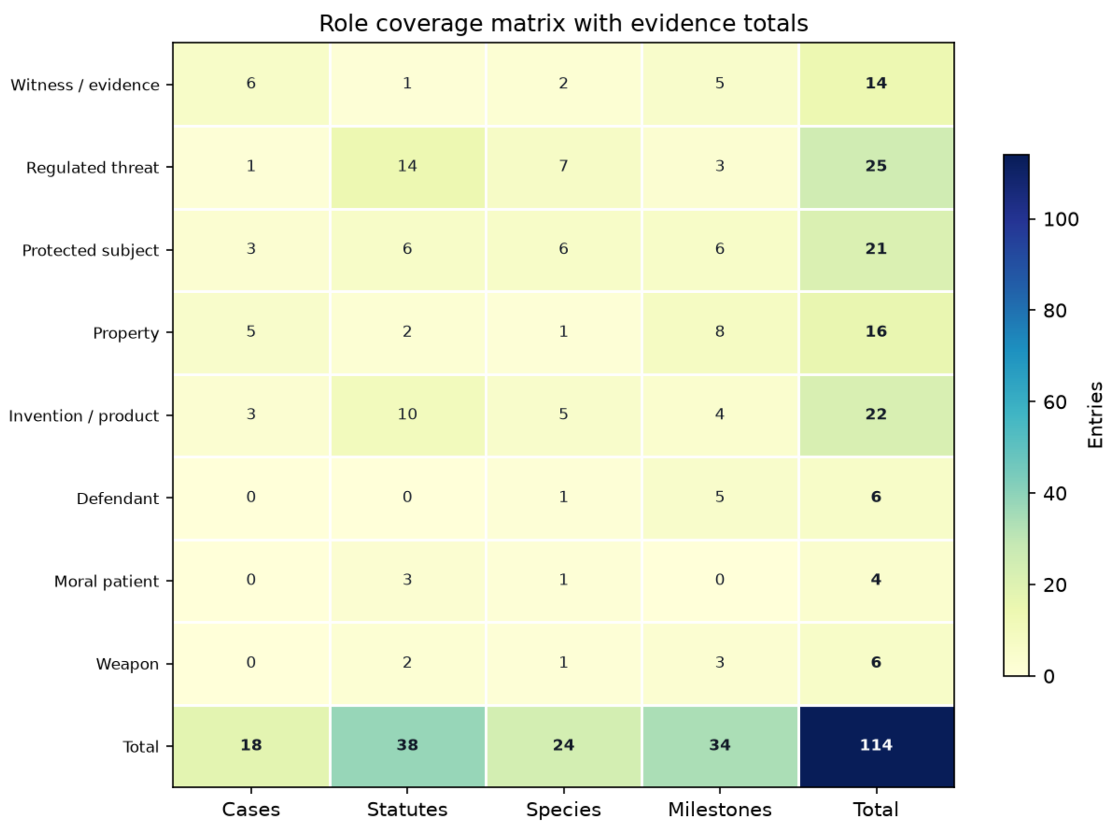

Figure 8: Heatmap of how much evidence each role carries across the four registry kinds (cases, statutes, species, milestones), making the field's case-driven, statute-driven, and history-driven corners visible at a glance. Read as: a role can be legally central even when its evidence is historical or statutory rather than case-law dense. Why it matters: the figure prevents case-counts from standing in for the whole legal architecture. Provenance: `src.metrics.role_coverage_matrix()`. Caveat: cell values are registry counts, not weights of legal importance.

before consumer-protection law can act on it [Utah Legislature, 2025]. In both settings, the legal question is not whether an insect exists; it is which institutional vocabulary makes the insect actionable.

The registry itself makes that classification work explicit. Each row asks what role the insect is playing, which institution is authorized to notice it, and what legal consequence follows from that notice. That mirrors scholarship on classification infrastructure and science/society co-production: the categories do not merely describe insects after law has acted; they help determine which facts can become legal facts in the first place [Bowker and Star, 1999, Jasanoff, 2004]. The insect becomes legally real when the right boundary is drawn around the right expertise, whether that boundary is a Daubert hearing, a quarantine perimeter, a listing rule, a patent claim, or a welfare threshold [Gieryn, 1983].

12 Methods: Registry-First, Claim-Ledgered Legal Synthesis

This reference is built so that the map and the evidence cannot drift apart. All domain knowledge lives in 7 source-of-truth registries under `src/` — roles, cases, statutes, species, institutions, timeline, and interconnections — and every reader-facing artifact is regenerated from them, as shown in the architecture figure.

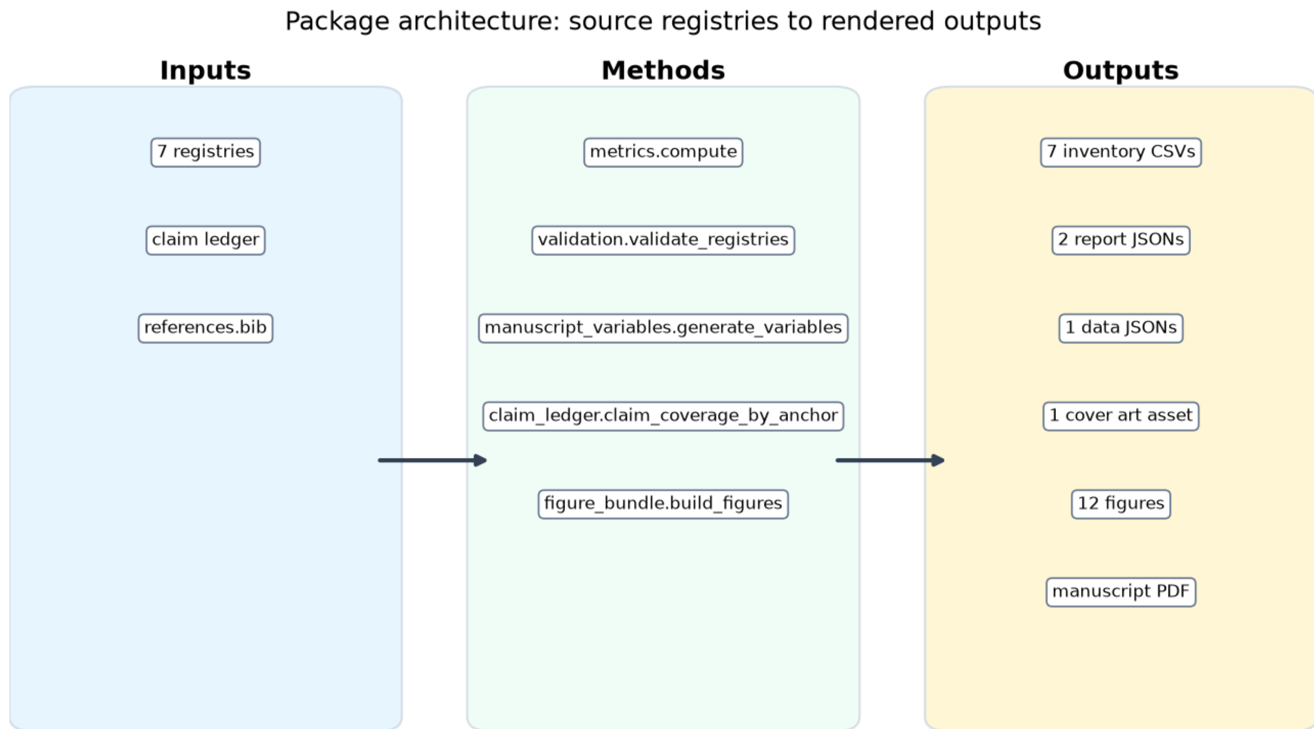

Figure 9: Data flow from the source-owned registries through pure generator methods to reproducible outputs: 7 registries feed metrics, validation, manuscript-variable generation, and figure rendering, which produce inventories, reports, figures, and the rendered manuscript. Read as: the package treats the manuscript as a compiled artifact, not as the source of legal facts. Why it matters: readers can audit whether prose, visuals, and counts share the same inputs. Provenance: `src/package_map.py`. Caveat: a local build-pipeline description, not a deployment diagram.

12.1 Token closure: every count comes from code

Every magnitude-bearing number in this prose is a double-brace token (a `TOKEN` wrapped in doubled braces) resolved at build time by `src.manuscript_variables.generate_variables`. The closure test asserts that every token referenced in the manuscript is generated and that the generator emits no orphan, so a registry edit that changes a count (say, adding a case) updates the prose automatically, and a hand-typed count that drifts from its registry fails the build. The 38 statutes span 7 jurisdictions, shown in the statutes-by-jurisdiction figure; that number, like the 18 cases and the 12 figures, is generated, never typed.

12.2 Claim ledger: volatile facts need source quotes

Counts come from the registries; *external* and volatile current-status claims do not. Each externally-sourced statistic written as a numeral, and each date-sensitive status claim that the manuscript needs to preserve, is registered in `data/claim_ledger.yaml` with a verification record: the source URL, a verbatim supporting quote, an as-of date, a confidence label, and the date the check was last run. The current ledger contains 20 quote-backed entries, with section coverage shown in the claim-ledger coverage figure. Two oracles bind these claims. The **offline** oracle (`src.claim_ledger.validate_claim_ledger`) proves each claim is attributed to a real bibliography key, anchored to a declared section, and carries a complete, fail-closed verification block — it cannot prove a number is *true*. The **live** oracle (`tests/test_live_claim_sources.py`, run with `-m live`) fetches each source URL and confirms the recorded quotes appear, so "verified" is a re-runnable fact rather than an assertion. This boundary is deliberate and documented: the green offline gate guarantees *shape and*

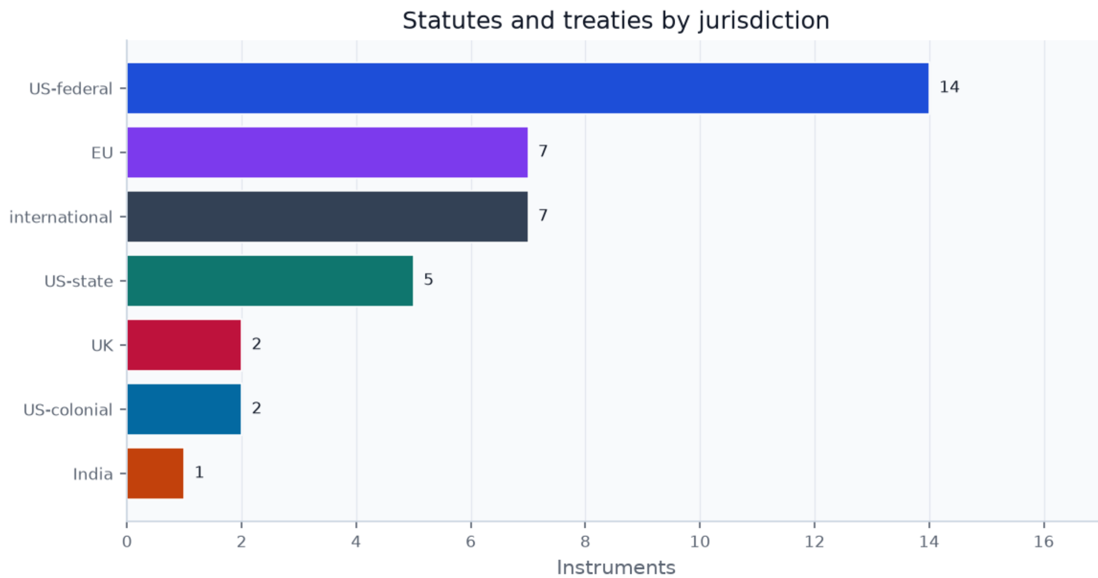

Figure 10: The same 38 instruments partitioned by issuing jurisdiction — US-federal, US-state, US-colonial, UK, EU, and international — showing the field's multi-level legal architecture. Read as: insect law is built through stacked authority, with local movement rules, national statutes, and international instruments all operating at once. Why it matters: no single legal layer owns the field's risk decisions. Provenance: `src/statutes.py`. Caveat: jurisdiction is the issuing level, not a measure of reach or enforcement.

attribution; only the live oracle guarantees *correspondence to the source*.

12.3 Validation gates for citations, cross-references, and statistics

A cross-registry validator fails closed on a malformed citation, an out-of-vocabulary role or category, a duplicate slug, or an empty required field, and writes its findings to `output/reports/validation.json`. A separate gate scans the manuscript for any numeral-form magnitude that is neither a generated token nor a value present in the claim ledger, with a negative control proving the detector fires on a planted statistic. Together these gates make the reference's central promise machine-checkable: no numeral-form statistic reaches the page unbacked.

The bibliography is also treated as data. The citation-date figure parses `references.bib` directly, so the reader can see where the source base is early legal text, where it is modern scholarship, and where it is official current-status material.

12.4 Reproducibility metadata and render provenance

This build was generated on `Darwin arm64` under Python `3.12.13` from configuration hash `a10accdf96648136` at `2026-07-01T23:58:18Z`. The same version-controlled inputs regenerate the inventories, the validation report, the figures, the manuscript variables, and this document with identical registry-derived content, modulo the provenance stamp.

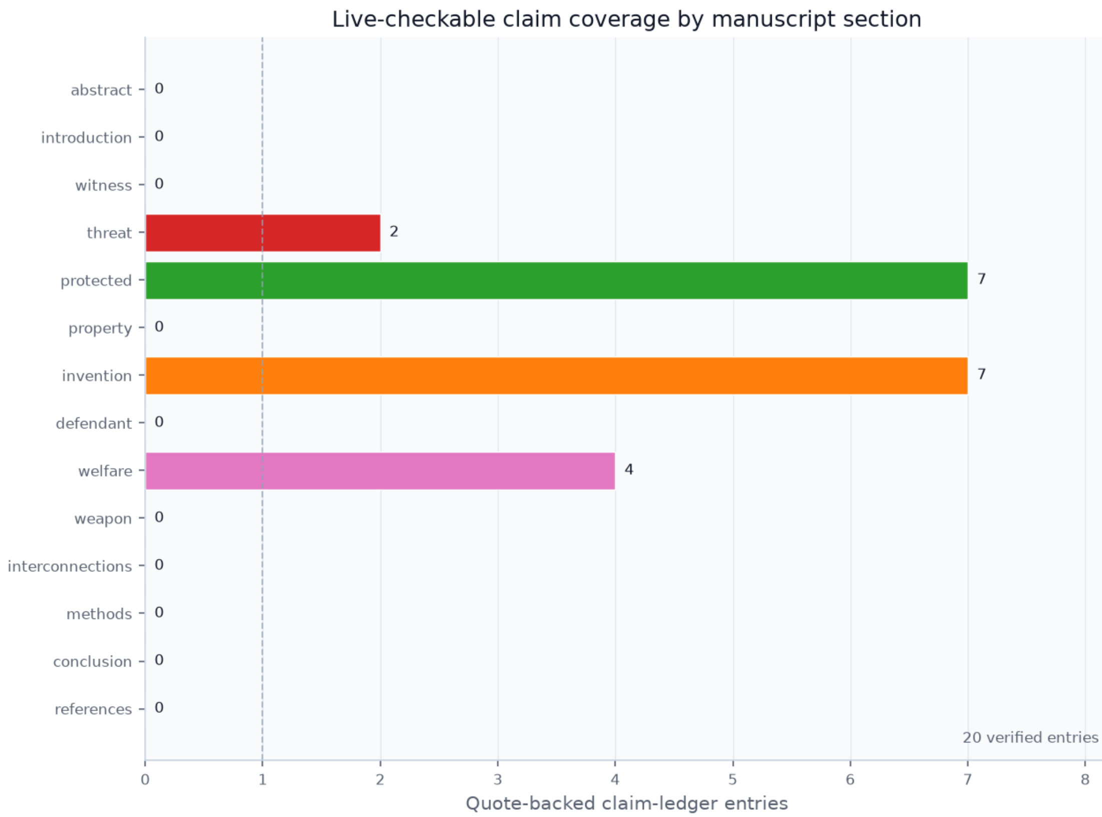

Figure 11: Coverage of the 20 live-checkable claim-ledger entries by declared manuscript section. Provenance: `src.claim_ledger.claim_coverage_by_anchor()`. Read as: volatile current-status and external magnitude claims are isolated where live evidence, not registry counts, carries the truth burden. Why it matters: it makes the manuscript's fact-checking boundary auditable by section. Caveat: a section with zero entries may still contain registry-derived facts or qualitative cited claims; this figure shows external/current claims that require a quote-backed verification block.

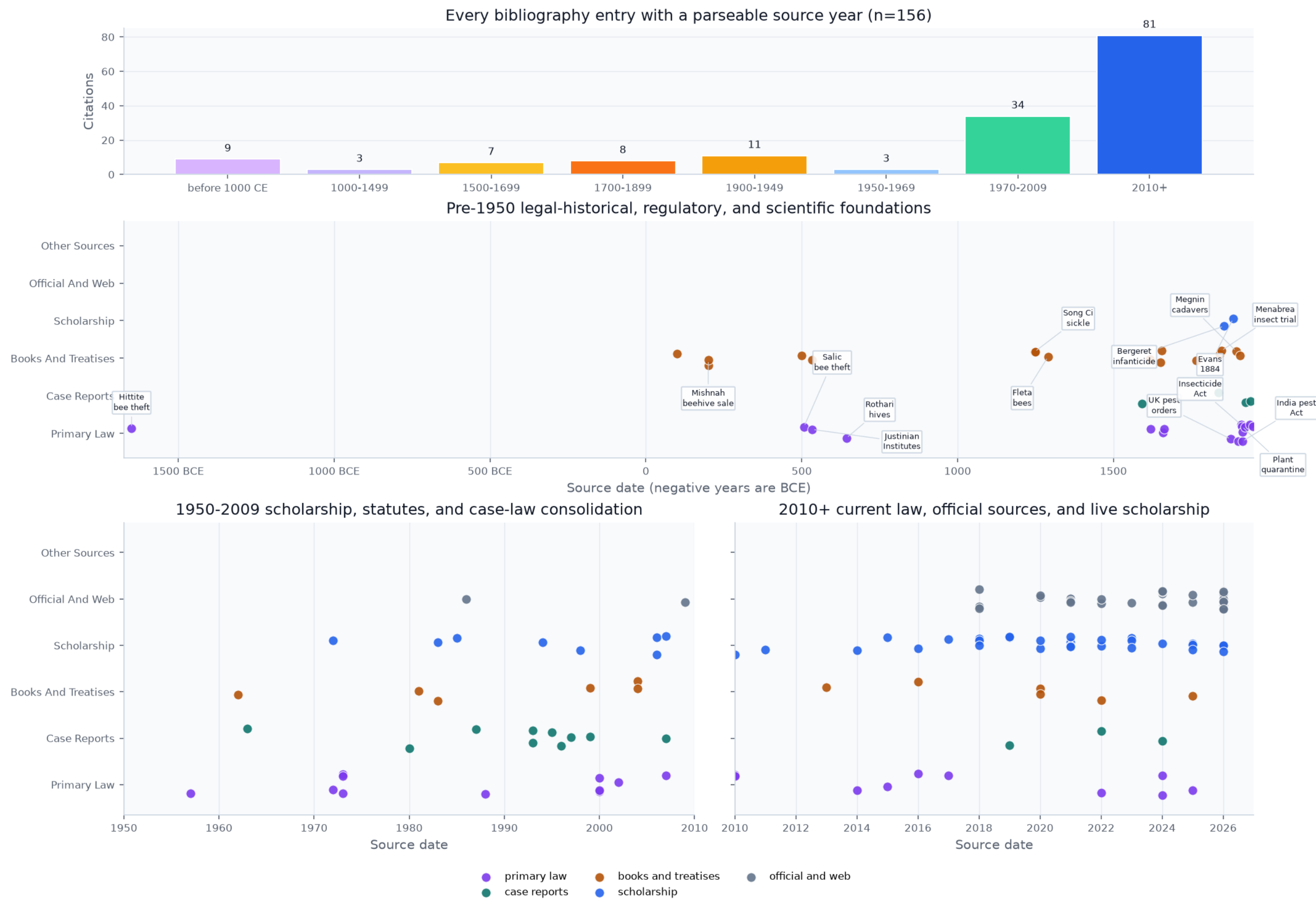

Figure 12: Date distribution for every bibliography entry with a parseable year, split into broad bands and individual source-date strips with the pre-1950 layer separated from later consolidation. Read as: EntoLaw's evidence base is anchored by early legal, regulatory, and treatise sources but interpreted through modern scholarship, cases, statutes, and official materials. Why it matters: the figure makes the historical depth of the citation stack visible instead of leaving it implicit in the reference list. Provenance: `manuscript/references.bib` parsed by `src.viz_citation_dates`. Caveat: the date is the bibliography year, so modern editions of older texts appear at their declared source date when that is how the project cites them.

13 Conclusion: Entomological Law as a Field Map

Entomological law is a genuinely transdisciplinary field whose coherence emerges not from a single statute or agency but from the biological ubiquity of insects themselves. Across the 8 roles mapped here, insects enter legal systems as evidence, as objects of protection or control, as property, as commodities and inventions, as historical defendants, as emerging welfare subjects, and as weapons — generating obligations and rights across many human–insect interactions, and tied together by 5 recurring legal devices.

Several frontiers deserve sustained attention. The **sentience question** — whether and when insects enter animal-welfare law — will likely be shaped by accumulating neuroscience, precautionary ethics, and legislative decisions in the EU and UK [Crump et al., 2023, Mikhalevich and Powell, 2020, de Souza Valente, 2025]. The **gene-drive governance question** is a domain where deployment-specific insect governance remains thin relative to the power of the technology [Convention on Biological Diversity, 2000, National Academies of Sciences, Engineering, and Medicine, 2016, Fisher, 2018, James et al., 2023]. And the **forensic-standardization problem** — divergent admissibility standards and uneven uptake of harmonized collection, preservation, and reporting practices — is a scientific–legal coordination failure with direct consequences for criminal justice [Amendt et al., 2007, Organization of Scientific Area Committees for Forensic Science, 2025].

The common frontier is institutional translation. Entomological facts do not enter law raw: they are filtered through admissibility rules, quarantine thresholds, listing standards, market-authorization pathways, and moral-status tests. That makes this field a laboratory for studying how legal systems convert nonhuman life into reviewable categories without erasing biological uncertainty [Jasanoff, 2015, Lodge et al., 2016, Lugo, 2006].

That is why the field deserves a shared name. "Entomological law" is not a claim that insects should have one code. It is a claim that a recurring classificatory transaction has become visible across many codes: insects become legal actors when scientific expertise, institutional jurisdiction, and social value line up enough to support action. The same transaction appears when possession turns a swarm into property, ecosystem services turn a fly or bee into a protected subject, and sentience evidence turns a cricket or fruit fly into a candidate for moral consideration [Rose, 1985, Losey and Vaughan, 2006, Bowker and Star, 1999].

The contribution of this reference is not a new doctrine but a new *substrate*: a registry-first, claim-sourced map in which every count is regenerated from source, legal propositions are source-bound in the bibliography, and every externally-sourced numeral is bound to a quotable, re-checkable source. The field has lacked its own treatise; what it most needs first is a machine-readable spine that future scholarship can extend without the map and the territory drifting apart. That spine is what this project supplies, and sec. 12 documents exactly how far its guarantees reach — and where they stop.

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